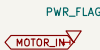
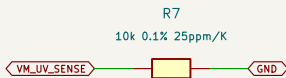
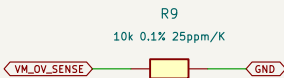
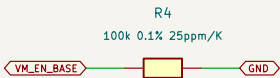
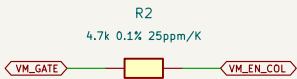
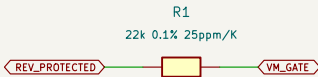
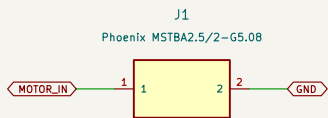
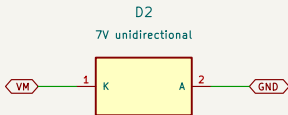
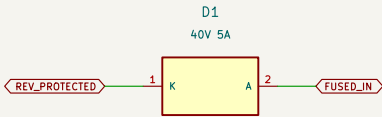
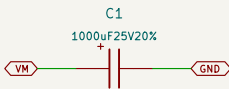
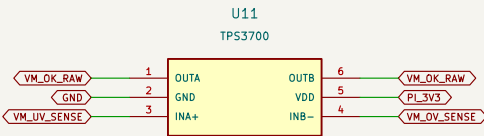
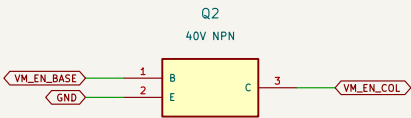
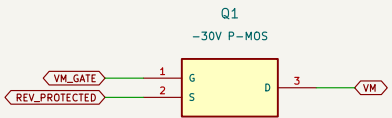
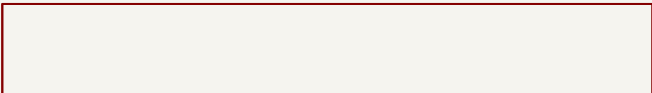


Power entry and rail qualifcation

5.5–6.5 V reference source; initial charge <=0.10 A, then <=3 A ceiling. Arbitrary battery hot–plug excluded.
PWR_FLAG marks intended external/conditional source connectivity only; it does not certify rail voltage or sequencing.

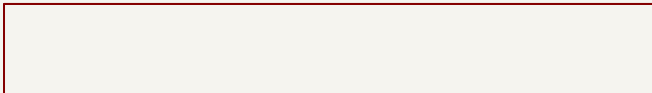


Interface



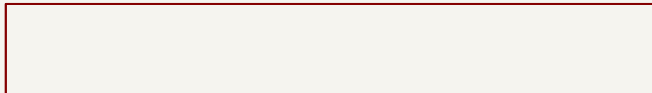
File: interface.kicad_sch

Safety



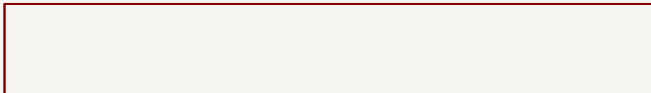
File: supervision.kicad_sch

Latch



File: enable_latch.kicad_sch

Drivers



File: motor_drivers.kicad_sch

PROVISIONAL – NOT VERIFIED FOR THE EXISTING ROBOT
NOT BUILT / NOT PHYSICALLY TESTED

Design only / pre–fabrication

Sheet: /
File: ai_robot_interface.kicad_sch

Title: **AI Robot Rev A / Power**

Size: A3 Date: 2026–09–28

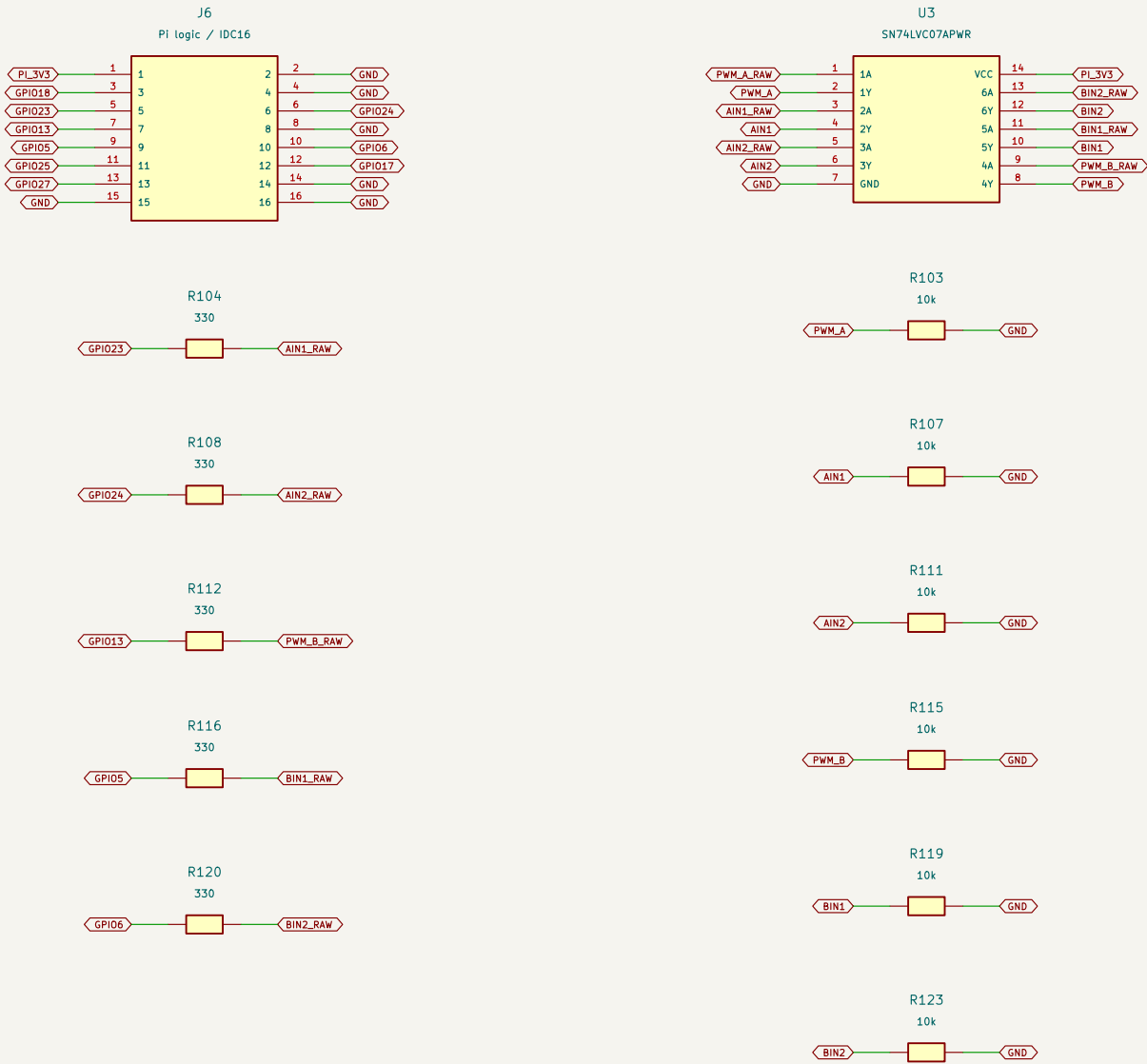
KiCad E.D.A. 10.0.6

Rev: **A–draft**

Id: 1/5

Pi logic interface / six buffered drive signals

Custom 16-way logic cable: NOT the Pi 40-pin header. Actual robot cable/GPIO correspondence remains unverified.
Open-drain buffer outputs use explicit pull-ups and pull-downs. Six logic nets fan out to U1/U2; motor outputs remain separate.



PROVISIONAL – NOT VERIFIED FOR THE EXISTING ROBOT
NOT BUILT / NOT PHYSICALLY TESTED
Design only / pre-fabrication

Sheet: /Interface/
File: interface.kicad_sch

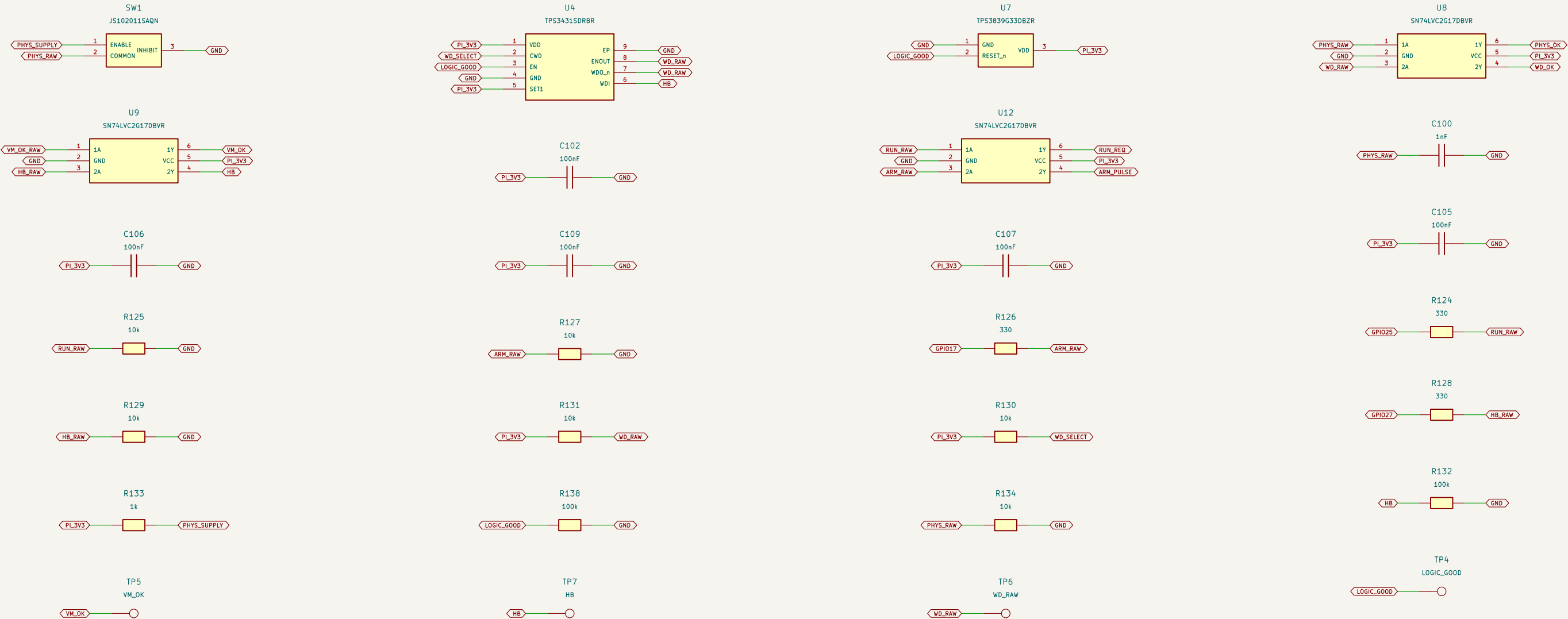
Title: AI Robot Rev A / Interface

Size: A3
Date: 2026-09-28
KiCad E.D.A. 10.0.6

Rev: A-draft
Id: 2/5

Supervision, manual disable and input conditioning

Independent watchdog, supply supervision and physical disable feed the enable logic; physical timing/behavior NOT TESTED.
GPIO17 is ARM pulse; GPIO27 is heartbeat; GPIO25 is RUN request. Qualified logic rails required before deliberate arming.



PROVISIONAL – NOT VERIFIED FOR THE EXISTING ROBOT
NOT BUILT / NOT PHYSICALLY TESTED

Design only / pre-fabrication

Sheet: /Safety/
File: supervision.kicad_sch

Title: AI Robot Rev A / Safety

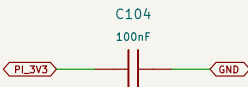
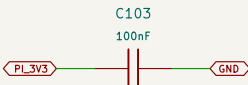
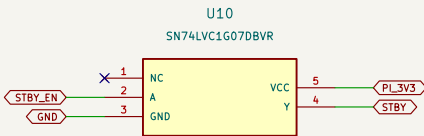
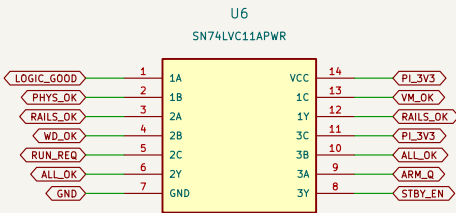
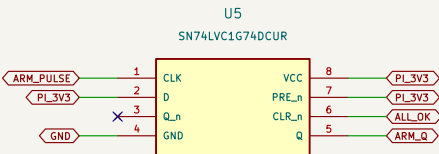
Size: A3 Date: 2026-09-28
KiCad E.D.A. 10.0.6

Rev: A-draft
Id: 3/5

Deliberate arm latch and hardware enable chain

Loss of ALL_OK asynchronously clears ARM_Q. A fresh qualified ARM_PULSE edge is required after recovery.

U10 open-drain output controls STBY with pull-up and two driver-local pull-downs; no output GPIO is shorted by SW1.



PROVISIONAL – NOT VERIFIED FOR THE EXISTING ROBOT
NOT BUILT / NOT PHYSICALLY TESTED

Design only / pre-fabrication

Sheet: /Latch/
File: enable_latch.kicad_sch

Title: **AI Robot Rev A / Latch**

Size: A3
Date: 2026-09-28

Rev: **A-draft**
Id: 4/5

Four separate motor channels / local decoupling

U1/Board A: front-left A and front-right B. U2/Board B: back-left A and back-right B. One motor per bridge.

Per motor reference envelope: <=0.25 A RMS; <=0.60 A peak for <=20 ms at <=10% duty. Lead polarity still unverified.

Design only / pre-fabrication

Title: AI Robot Rev A / Drivers

	7	8
--	---	---